Where a Dark-Energy-Tied Galaxy Law Holds and Where It Breaks: Low-Acceleration Tests, a Two-Footing Shear Pass, and the Satellite-Dwarf Excess

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Abstract

We report one day of pre-registered tests of a gravity framework whose galactic law is $g_N = g^2/(g + a_0)$ with $a_0 = \frac{1}{2}c\sqrt{G\rho_{\rm DE}}$ (the $\frac{1}{2}$ is fitted). Its extra gravitating mass is a collisionless "cold energy" component. Every lane froze its criteria before computing and ran a mutation control; both a_0 footings (9.36×10^{-11} and $1.13 \times 10^{-10}\,\mathrm{m\,s^{-2}}$) are reported and never pooled.

1. **Shear.** The framework's own 1024^3 particle-mesh structure, scored against raw KiDS-1000 cosmic shear, is CONSISTENT on both footings ($\Delta\chi^2 = -7.70$ and -7.18 against its matched control). The pass is conditional on inherited initial conditions; a native-input rerun is running.
2. **Where the law holds.** Isolated late-type galaxies obey the law down to $g_{\rm bar} \approx 7 \times 10^{-14}\,\mathrm{m\,s^{-2}}$ in KiDS-1000 lensing (amplitude 1.016 ± 0.086; within $\sim$0.1 dex across robustness variants).
3. **Where it breaks.** Spheroidal / red isolated lenses sit 1.45× above the law at the same accelerations after a stellar-mass correction (4.0σ split). They follow $g_{\rm obs} = g_{\rm law}(g_\dagger/g_{\rm bar})^{0.060}$. Milky Way and M31 satellite dwarfs carry a larger, host-independent excess ($\approx +0.44$ dex) that ten framework-native mechanisms fail to produce.
4. **Theory.** Lensing equals dynamics in the framework to better than 2×10^{-5}. In the candidate particle model a uniform field produces no internal response at all, so the law's cold mass must be real mass assembled from afar, not a local polarisation.

1 The shear test on both footings

The framework's combined "theory-v2" particle-mesh engine (per-host supply, condensed source, gas expulsion; 1024^3 particles, $100\,\mathrm{Mpc}/h$) was scored by an unchanged pipeline on raw KiDS-1000 and DES Y3 $\xi_\pm$ against a matched control run.

footing	KiDS-1000 $\Delta\chi^2$	robustness range	DES Y3
canonical	-7.70 (CONSISTENT)	-4.6 to -11.0	not diagnostic
alternative	-7.18 (CONSISTENT)	-4.3 to -10.2	not diagnostic

The frozen rule requires both footings, so the shear side passes. It is conditional on inherited ΛCDM-shaped initial conditions and background, a post-preview module selection and a post-hoc control, all disclosed. Runs with the framework's own initial conditions are in progress and decide the condition.

2 Isolated galaxies obey the law at ultra-low acceleration

- **The test.** We computed the full-covariance amplitude A of the Brouwer et al. (2021) isolated-lens radial acceleration relation over the law, in the window $10^{-13.3} \le g_{\rm bar} < 10^{-11.9}\,\mathrm{m\,s^{-2}}$, which is where the satellite dwarfs sit.
- **Late types.** Blue (late-type) lenses give $A = 1.016 \pm 0.086$, excluding the dwarfs' $\times 2.75$ at 20.2σ.
- **Robustness.** Window shifts, a ± 0.2 dex stellar-mass systematic, a Sérsic definition and the alternative footing move A between 0.902 and 1.312. A frozen robustness lane therefore classes "exactly on the law" as fragile, but the dwarf excess stays excluded by $\ge 12.2\sigma$ in every variant.
- **Scope.** Lensing probes 30–300 kpc. This tests the law's $g(g_{\rm bar})$ relation, not extra mass concentrated inside small systems.

3 The spheroidal excess

- **Both splits.** At the same accelerations, red and spheroidal ($n > 2$) lenses sit $\sim 1.6\times$ above the law, and blue and disc-like lenses on it. The colour split is 5.0σ and the Sérsic split 4.6σ; the two labels cannot be separated with the published bins.
- **Stellar-mass correction.** Applying the measured $+0.123$ dex offset between LePhare and colour-based masses for old early types leaves the split at 4.0σ (red $A = 1.451$). With $+0.20$ dex it is 2.9σ.
- **An empirical law.** Red lenses follow $g_{\rm obs} = g_{\rm law}(g_{\rm bar})\,(g_\dagger/g_{\rm bar})^b$ with $b = 0.060$ (0.050–0.070) and $\log_{10} g_\dagger = -10.05$. This gives $\chi^2 = 16.1$ against 157.9 for the law alone on 15 points; it is a description, not a prediction.
  - With corrected masses on the canonical footing the onset lies near a_0. This is footing-dependent and is flagged as a coincidence to test.
- **Not one law with the dwarfs.** Extrapolated to the satellite dwarfs, the same law predicts $+0.147$ dex against an observed $+0.438$ (4.4σ low).

4 The satellite-dwarf excess and what does not explain it

The excess. With stellar masses from each dwarf's own star-formation history (E-MILES $\times$ Weisz et al. 2014), 25 of 28 independent dwarfs sit above the law, with mean $\ln(\sigma_{\rm obs}/\sigma_{\rm law}) = +0.38$. There is no age dependence (Spearman $\rho = +0.09$); an apparent age trend at fixed $\Upsilon = 2$ was a stellar mass-to-light artefact.

- It lives only at $g_{\rm bar} < 10^{-12}$. Where dwarfs overlap SPARC discs they agree ($+0.05 \pm 0.07$ dex).
- It is the same for MW and M31 satellites at matched acceleration (offset $+0.055 \pm 0.117$ dex).

Mechanisms excluded, each by a frozen, mutation-controlled test:

- stiffening of gravity: globulars need $L_* \geq 24$ pc, dSphs ≤ 22 pc;
- free cold energy of expelled gas spread to the turnaround radius (too far out);
- the catchment budget packed into $r_{1/2}$: tight at $\Upsilon = 2$ (median 0.30 of the whole supply), failing at $\Upsilon = 1$;
- present-day tides ($\rho = -0.00$) and pericentric tides from Gaia orbits in the framework's own Milky Way field ($\rho = +0.08$);
- capture of the host's cold energy: the Liouville ceiling is 10^{-8}–2×10^{-5} of the need;
- an early-collapse free core in every small system: it reaches the densities but over-predicts SPARC's small rotating dwarfs ($+0.305$ against $+0.032$ dex, 10.4σ);
- maximum-entropy relaxation of the two-population cold medium: wrong sign or ill-posed;
- gas loss traced by metallicity deficit ($\rho = -0.11$, with power shown by injection);
- dynamical-friction transport by fragmented cold energy: it over-fills Fornax $\times 6.0$ and Leo I $\times 5.8$.

The constraint any mechanism must meet. Fainter dwarfs need more absolute extra mass at fixed radius and dispersion. Leo I has Draco's size and dispersion and six times its stars, but needs less extra mass.

5 Theory results

- **Lensing equals dynamics.** For parents of the record's type (dust-like cold energy plus the stiffened field), the gravitational slip $|\Phi - \Psi|/v_c^2$ is $\leq 1.9 \times 10^{-5}$ from dwarfs to clusters on both footings, far below lensing precision. This follows largely from the dust-like premise. The field's own energy and stiffness are smaller still. A well-posed covariant parent is not shown.
- **No response to a uniform field.** In the candidate particle model, a uniform acceleration produces only a rigid translation of the total centre of mass: no internal change at any order, for any state, with host and recoil included.
  - A local response can generate at most $\approx 3\ell/r$ of the law's medium mass, for carriers displaced by ℓ.
  - The cold mass must therefore be real mass assembled from afar, which independently supports the catchment-supply rule.
  - The first missing selection equation is a medium kernel $4\pi G\Sigma_{\rm med} = a_0 g/(a_0 + g)$ across the galactic band.
- **Kinetic confinement.** For the two-population cold medium, the full linear feedback work is bounded uniformly in time if there is no embedded neutral mode and the orbital frequencies are non-degenerate (a conditional theorem). The energy method alone cannot close it, because the operator's continuous spectrum

covers every receiving frequency. Unconditional all-time confinement remains open.
- **Shape test.** A calibration-free shape test (from an independent AI-assisted review) between the simple law and a quadratic rival is not diagnostic on SPARC, because the gas-dominated discs lack transition-regime leverage.

6 What this establishes

- **What holds:** rotating and gas-rich systems obey the law at every acceleration tested; the shear side passes conditionally on both footings; lensing traces dynamics.
- **What is open:** spheroidal systems carry a slow excess below $\sim a_0$, and satellite dwarfs a larger one that no framework mechanism yet produces.
- **What is not derived:** $\kappa = \frac{1}{2}$, the cold-to-baryon supply ratio and the retained fraction remain inputs.

The sealed external tests (ALMA, Gaia DR4) are unaffected.

Public repository lanes (frozen criteria, scripts, outputs): CFG650 (shear), CFG820–CFG853 (dwarfs, lensing, mechanisms), CFG827 (SFH-based M/L), CFG829/840/849/850/851 (lensing), CFG847 (slip), and the acceleration-response checkpoint CL-AR-2026-10-19-01. Data: M. M. Brouwer et al., A&A 650, A113 (2021); F. Lelli, S. McGaugh, J. Schombert, AJ 152, 157 (2016) (SPARC); D. R. Weisz et al., ApJ 789, 147 (2014); A. Vazdekis et al. (E-MILES SSP models); the Local Volume Database; Arroyo-Polonio et al. (2026) (binary-corrected UFD dispersions); Gaia proper motions as compiled there. AI assistance: Anthropic's Claude, with an independent AI-assisted analytic review.